

Processing Big Data

2025/2026

Project Description

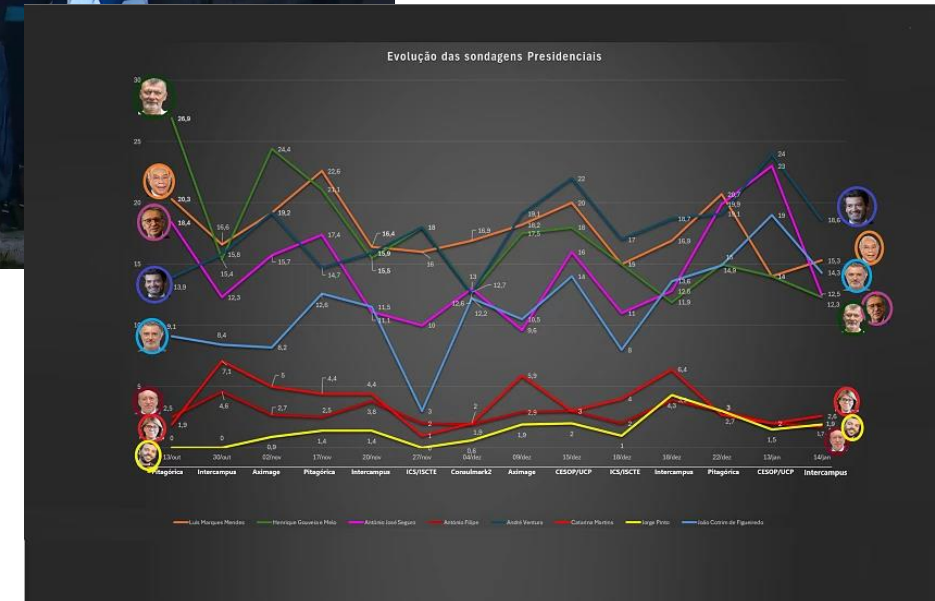
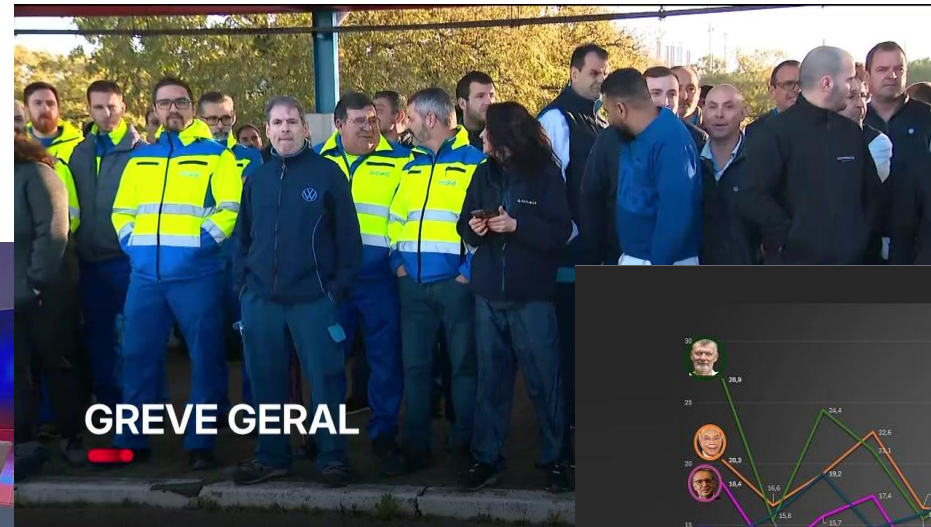
- Data context:
 - videos of 1-on-1 2025 debates between the main presidential candidates (28 videos total);
 - 7 newscast from 2 tv channels;
 - timeline of poll evolution.

Project Description

General Data Processing

- 1 frame per second for each video
- Audio diarization – split into segments with a single speaker
- Audio2Speech – Transcripts
- Additional text features (polls)

Multimodal Data!



▶ 0:00 / 0:26 — 🔊 ⋮

a candidatura confirma desta forma que é um forte candidato nas eleições presidenciais. Ele quer ser líder do opcional, quer reafirmar isso com gestos que quer substituir e, portanto, primeiro neste Luís Montego. Ou seja, está a preparar as eleições legislativas. É um forte candidato aqui, porque tentou arranjar candidatos, não conseguiu malter de nativa em condições, mas está destinado a enganar os eleitores.

Project Description

Visual Features - Debates

- **detected humans** – bounding boxes + class+confidence.
- **detected human poses** – 2D coordinates of 17 keypoints for each detected human (plus: probability of being visible), in a matrix of 17 x 3 for each pose.
- **detected faces**
 - **location** of each detected face, [x1, y1, x2, y2] (two corners of a bounding box);
 - **top emotion** given by a string, with up to 8 possible classes;
 - **Confidence** given by a probability;
 - **facial landmarks** – 2D positions of 106 keypoints in the face.

Project Description

Audio & Speech Features - Debates

- **pitch, harmonic to noise ratio (HNR), jitter, shimmer, formants, and formants-based features** - each represented by a scalar value;
- **speech-rate features** – number of pauses, speaking-rate, articulation-rate, speaking time;
- **speaker embedding vector** of dimension 512 extracted from the pyannote diarization deep learning model;
- **transcript** of the audio performed by Whisper;
- **transcript embedding** computed using EuroLLM.

Project Description

Visual and Speech Features - Newscast

- **OCR** – text location and recognition in the video frames:
 - **location** – Bboxes with text positions;
 - **text** – mapping between detected characters and Portuguese words;
 - **confidence** – associated to the predicted text.
- **transcript** of the audio performed by Whisper;
- **transcript embedding** computed using EuroLLM.

Project Description

Text Features

- **Polls** – evolution of pools over time;
- **Candidate2Party** – mapping between the candidates and the supporting parties.

Project Objectives

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1. Exploratory Data Analysis

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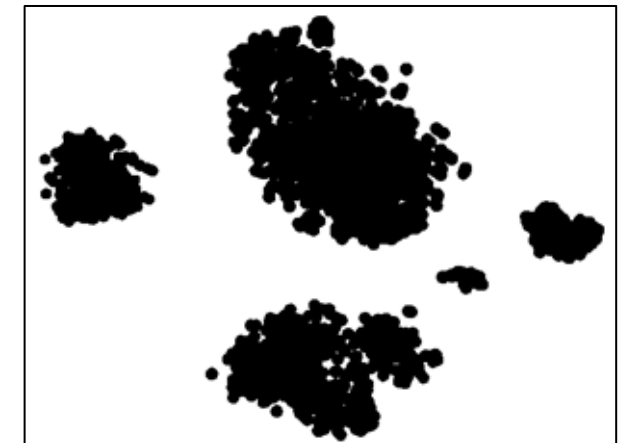
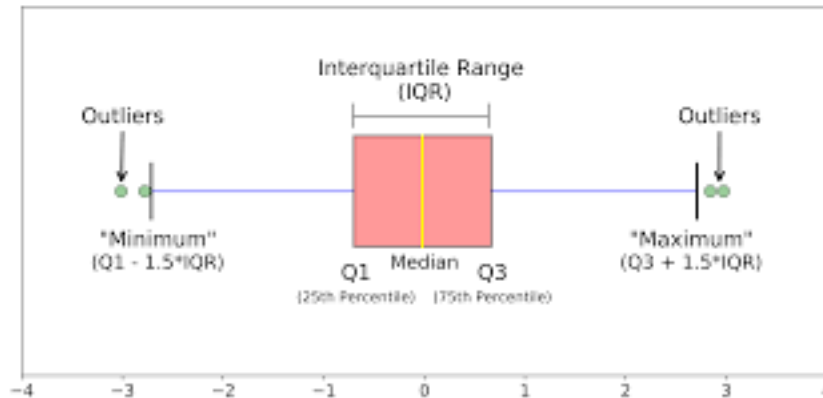
1. Exploratory Data Analysis

Data statistics

Histograms

2D Visualization of
features

Box Plots

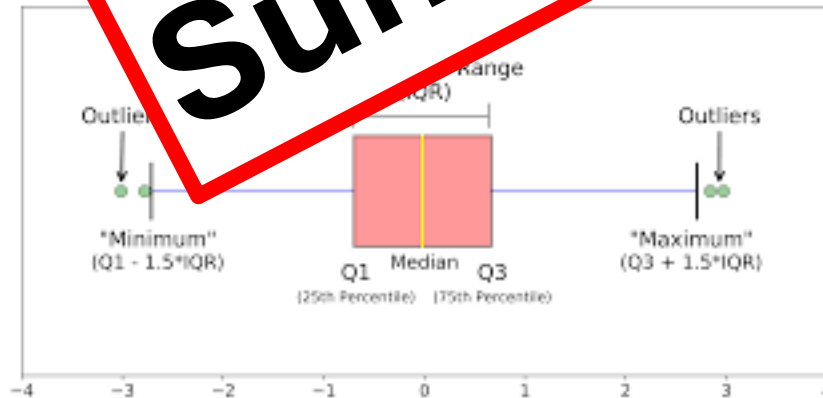


Project Objectives

1. Exploratory Data Analysis

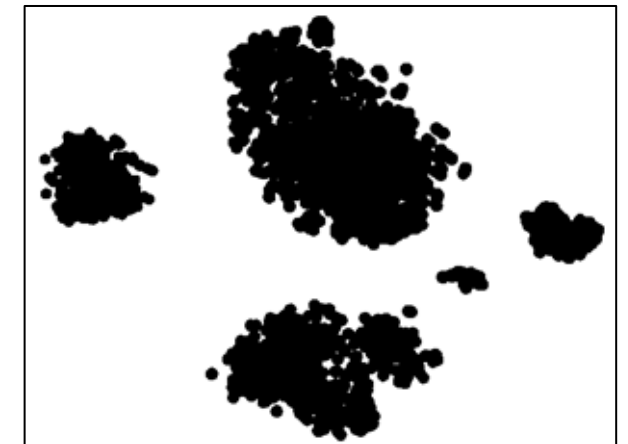
Data statistics

Box Plots



Sunday: May 10th

2D Visualization of features



Project Objectives

1. Exploratory Data Analysis
2. Single Video Analysis

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A. Newscast:

- A. Identify transitions between the anchor and news;
- B. Identify different topics, count the number of different news, and their length.

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B. Debates

- A. Segment the debate into moments;
- B. “Identify” people across frames and audio segments;
- C. Try to find correlations between features;

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1. Exploratory Data Analysis

2. Single Video Analysis

A. Newscast:

- A. Identify transitions between the anchor and news;
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B. Debates

- A. Segment the debate into moments;
- B. “Identify” people across frames and audio segments;
- C. Try to find correlations between features;

C. Is one modality enough to achieve the goals?

Project Objectives

1. Exploratory Data Analysis
2. Single Video Analysis
3. Multiple Video Analysis
 - A. ?
 - B. ?
 - C. ...

D. The sky is the limit! What do the data tell you?

Own the project and pick the direction(s) that interest you the most!

Project Challenges

- **Multimodal data!**
- **Different time stamps** – Frames and Audio/Speech are not 100% aligned
- **Missing** – missing features in:
 - Visual (e.g, missing keypoints);
 - Audio & Speech
- **Noise in the features**
- **Different feature dimensions**

Project Evaluation

- Groups of 4 (“by the end of week 1 incomplete groups will be reshuffled to attain full capacity”)
- Project is worth 65% of the final grade (min: 9.5):
 - Continuous evaluation during PB classes 10%
 - Midterm review – **week 4** – 8min presentation + 7min discussion about Task 1 (**Slide deck on Fénix by Sunday May 10th**) 15%
 - Code – submit by **June 2nd** 20%
 - Video report – submit by **June 6th** 20%
 - Discussion (20min) – **week 8**

Let's look at the data!

- Use any coding language you want, but I only “speak” python

Important python libraries

- Pandas <- useful to manage and manipulate (big) data
- Numpy <- useful .
- Sklearn <- will do most of the work for you
- Matplotlib <- useful visualization tool
- Jupyter <- nice and practical interface